

Satyajit Puhan

Postdoctoral Researcher in High Energy Physics

Institute of Physics, Academia Sinica – Taipei, Taiwan

☎ +91-9776342407 • ✉ puhansatyajit@gmail.com • 🌐 satyajitpuhan.github.io

🆔 0009-0004-9766-5005 | 🎓 Google Scholar | ⚙️ InspireHEP | 📄 ResearchGate | 🔗 LinkedIn |
🌐 Website | 🐙 GitHub

Education

Degree	Institution	Score	Year
Post-Doc (HEP)	Institute of Physics, Academia Sinica, Taiwan	—	2026–Present
PhD (HEP)	Dr. B. R. Ambedkar NIT Jalandhar, Punjab	8.75/10	2022–2026
M.Sc (Physics)	NIT Jamshedpur, Jharkhand	8.59/10	2019–2021
B.Sc (Physics)	Bhadrak Autonomous College, Odisha	8.91/10	2016–2019
Higher Secondary (12 th)	Siddhivinayak +2 Science College, Balasore	78%	2015–2016
Higher Secondary (11 th)	FakirMohan Junior College, Balasore	—	2014–2015
High School	Harekrushna High School, Balasore	87%	2011–2014

Research Summary

Postdoctoral Research (2026–Present)

Institute of Physics, Academia Sinica, Taipei, Taiwan

Supervisor: Prof. Wen-Chen Chang

- **Meson Structure via the Sullivan Process:** Extraction of pion and kaon parton distribution functions (PDFs) within the **Sullivan process framework** using phenomenological and neural-network approaches. Studied leading-neutron electroproduction at HERA and the EIC, including target fragmentation and constraints on pion PDFs.
- **Detector Development for ePIC:** Working on the **Zero-Degree Calorimeter (ZDC)** for the ePIC experiment at the upcoming Electron-Ion Collider (EIC), Brookhaven National Laboratory, as part of the far-forward program essential for leading-neutron (Sullivan process) measurements.
- **Deep-Learning Extraction of GPDs:** Extraction of pion unpolarized quark and gluon generalized parton distributions (GPDs) using deep neural networks.

PhD Research (2022–2026)

Dr. B. R. Ambedkar NIT Jalandhar, Punjab

Supervisor: Dr. Harleen Dahiya

Thesis: *Tomography of Light and Heavy Mesons Using Light-Front Dynamics* (defended 2nd April 2026).

- **Transverse Momentum Dependent Distributions (TMDs):** Studied model-dependent TMDs for spin-0 (pion, kaon), spin- $\frac{1}{2}$ (proton), and spin-1 particles (rho meson, heavy vector mesons, photon) in light-front dynamics. Calculated associated PDFs and spin densities in momentum space. Extracted pion unpolarized TMDs from Drell–Yan data using neural-network methods.
- **Generalized Parton Distributions (GPDs):** Calculated GPDs for spin-0 (pion, kaon) and spin-1 (rho meson) particles; extracted corresponding form factors and initiated work on generalized distribution amplitudes (GDAs). Data-driven extraction of pion unpolarized quark GPDs from electromagnetic form factor data. Relevant for upcoming **Electron-Ion Collider (EIC)** and **Jefferson Lab (JLab)** experiments.
- **Physical Properties:** Computed spin-orbit correlations for mesons, distribution amplitudes (DAs), electromagnetic form factors, charge radii, decay constants, mass spectra, radially excited states of quarkonia, and radiative transitions of charmonia within the light-front quark model. Studied internal structure using power-law wave functions.

- **Mechanical Properties:** Calculated gravitational form factors (GFFs) of the pion, including pressure and shear-force distributions. Ongoing collaboration with **Prof. Oleg V. Teryaev (JINR)** on kaon GFFs and energy-momentum tensor calculations.
- **Medium Modifications:** Studied medium modification effects on TMDs, PDFs, and GPDs in nuclear and hadronic matter using light-front quark model, light-front holographic model, and chiral SU(3) quark mean-field model. Investigated pion, kaon, B-meson, and D-meson properties in asymmetric nuclear matter at finite temperature.
- **Extraction via Neural Networks:** Extracted pion TMDs and PDFs from Drell–Yan data using neural-network approaches. Working on unpolarized and helicity TMDs for the proton from E288/FNAL Drell–Yan data, and GPD extraction via EMFF data.
- **Transport Coefficients of Strongly Interacting Matter:** Calculating transport coefficients within the Nambu–Jona-Lasinio (NJL) model with Polyakov loop, in collaboration with Ashutosh Dwibedi and **Prof. Sabyasachi Ghosh (IIT Bhilai)**.
- **Nuclear Physics (Collaboration):** Calculated higher-order proton–oxygen scattering cross sections by interpreting the Yamaguchi nuclear potential in the Hamiltonian, in collaboration with **Prof. Ujjwal Laha**.

Collaborations & Working Groups

Electron-Ion Collider (EIC)

EIC Meson Structure Working Group

2026–Present

Physics studies of pion and kaon structure functions via the Sullivan process (leading-neutron tagging) at the EIC.

ePIC Collaboration, Electron-Ion Collider (EIC)

Zero-Degree Calorimeter (ZDC) Working Group

2026–Present

Detector development and simulation studies for the far-forward Zero-Degree Calorimeter.

Visiting Fellowships

JINR, Dubna

Russia

Visiting Fellow

Jun 16 – Sep 12, 2025

Under the guidance of **Prof. Oleg V. Teryaev**. Worked on GPD extraction using neural network methods and calculation of energy-momentum tensors (gravitational form factors).

Indian Institute of Technology Bhilai

Chhattisgarh

Visiting Fellow

Feb 16–23, 2025

Under the guidance of **Prof. Sabyasachi Ghosh**. Worked on applications of the NJL model to understand hadron structure and in-medium meson properties.

Publications

INSPIRE-HEP metrics (Sep 2026): 43 records | 212 citations | *h*-index: 10 | 23 peer-reviewed journal articles (1 single-author)

Peer-Reviewed Articles (Published)

- [1] **S. Puhan**, S. Sharma, N. Kaur, N. Kumar and H. Dahiya, “T-even TMDs for the spin-0 pseudo-scalar mesons up to twist-4 using light-front formalism,” *J. High Energy Phys.* **02** (2024) 075. [arXiv:2310.03464](#)
- [2] **S. Puhan** and H. Dahiya, “Leading twist T-even TMDs for the spin-1 heavy vector mesons,” *Phys. Rev. D* **109**, 034005 (2024). [arXiv:2310.03465](#)
- [3] R. Acharyya, **S. Puhan** and H. Dahiya, “Quark spin-orbit correlations in spin-0 and spin-1 mesons using the light-front quark model,” *Phys. Rev. D* **110**, 034020 (2024). [arXiv:2405.00446](#)
- [4] S. Sharma, **S. Puhan**, N. Kumar and H. Dahiya, “TMD relations: insights from a light-front quark-diquark model,” *Prog. Theor. Exp. Phys.* **2024**, 103B05 (2024). [arXiv:2405.13727](#)
- [5] **S. Puhan**, N. Kaur, A. Kumar, S. Dutt and H. Dahiya, “Pion valence quark distributions in asymmetric nuclear matter at finite temperature,” *Phys. Rev. D* **110**, 054042 (2024). [arXiv:2408.07334](#)
- [6] N. Kaur, **S. Puhan**, R. Pandey, A. Kumar, S. Dutt and H. Dahiya, “Does nuclear medium affect the transverse momentum-dependent parton distributions of valence quark of pions?” *Phys. Lett. B* **859**, 139114

(2024). [arXiv:2409.05394](#)

- [7] **S. Puhan**, N. Kumar and H. Dahiya, “Photon leading twist transverse momentum dependent parton distributions,” *Eur. Phys. J. A* **61**, 56 (2025) [Topical edition on Spin-1 observables]. [arXiv:2408.07714](#)
- [8] R. Acharyya, **S. Puhan**, H. Dahiya and N. Kumar, “Spectroscopy of excited quarkonium states in the light-front quark model,” *Chin. Phys. C* **49**, 023104 (2025). [arXiv:2408.07715](#)
- [9] **S. Puhan**, N. Kaur and H. Dahiya, “Transverse and spatial structure of light to heavy pseudoscalar mesons in light-cone quark model,” *Phys. Rev. D* **111**, 014008 (2025). [arXiv:2410.07596](#)
- [10] D. Singh, **S. Puhan**, N. Kaur, M. Kaur, A. Kumar, S. Dutt and H. Dahiya, “Effect of an asymmetric nuclear medium on the valence quark structure of the kaons,” *Phys. Rev. D* **111**, 054001 (2025). [arXiv:2410.20181](#)
- [11] **S. Puhan**, N. Kaur, A. Kumar, S. Dutt and H. Dahiya, “Effect of nuclear medium on the spatial distribution of pions,” *Nucl. Phys. B* **1017**, 116940 (2025). [arXiv:2501.16706](#)
- [12] **S. Puhan** and H. Dahiya, “Scalar, vector, and tensor form factors of pion and kaon,” *Phys. Rev. D* **111**, 114039 (2025). [arXiv:2505.02507](#)
- [13] **S. Puhan**, S. Sharma, N. Kumar and H. Dahiya, “Understanding the valence quark structure of the pion through generalized transverse momentum-dependent parton distributions,” *Prog. Theor. Exp. Phys.* **2025**, 083B02 (2025). [arXiv:2504.14982](#)
- [14] Tanisha, **S. Puhan**, A. Yadav and H. Dahiya, “Valence quark distribution of light ρ and heavy J/ψ vector mesons in a light-cone quark model,” *Phys. Rev. D* **112**, 054035 (2025). [arXiv:2505.09213](#)
- [15] R. Pandey, **S. Puhan**, N. Kaur, A. Kumar, S. Dutt and H. Dahiya, “Valence quark properties of charged kaons in symmetric nuclear matter,” *Eur. Phys. J. Plus* **140**, 1040 (2025). [arXiv:2505.01202](#)
- [16] A. Gautam, D. Singh, N. Kaur, **S. Puhan**, S. Dutt, H. Dahiya and A. Kumar, “ D and D^* mesons in an isospin asymmetric nuclear medium,” *Prog. Theor. Exp. Phys.* **2025**, 113D03 (2025). [arXiv:2506.08707](#)
- [17] Tanisha, **S. Puhan**, N. Kaur, H. Dahiya and A. Kumar, “Impact of isospin asymmetric nuclear medium on pseudoscalar and vector B mesons,” *Nucl. Phys. A* **1068**, 123330 (2026). [arXiv:2504.21392](#)
- [18] **S. Puhan**, S. Sharma, N. Kumar and H. Dahiya, “Valence quark distribution of the rho meson using a light-front quark model,” *Phys. Rev. D* **113**, 036030 (2026). [arXiv:2511.10981](#)
- [19] A. Gautam, Tanisha, **S. Puhan**, A. Kumar and H. Dahiya, “Spectroscopy of the ρ meson in a symmetric nuclear medium,” *Phys. Rev. D* **113**, 114043 (2026). [arXiv:2601.11082](#)
- [20] A. Dwibedi, **S. Puhan**, S. Ghosh and H. Dahiya, “Valence quark distribution of the pion inside a medium with finite baryon density: A Nambu–Jona-Lasinio model approach,” *Prog. Theor. Exp. Phys.* **2026**, 063B04 (2026). [arXiv:2512.24921](#)
- [21] Hari Govind P, **S. Puhan**, Abhishek K. P, R. Pandey, H. Dahiya, A. Kumar and S. Dutt, “Pion parton distribution functions in the light-cone quark model and experimental constraints,” *Phys. Rev. D* **114**, 016010 (2026). [arXiv:2604.05762](#)
- [22] **S. Puhan**, N. Kumar and H. Dahiya, “Internal structure of light mesons using the power law wave function,” *Mod. Phys. Lett. A* **41**, 2642011 (2026) [Special Issue]. [arXiv:2604.07049](#)
- [23] **S. Puhan** (single author), “Analysis on the parton distribution functions of heavy mesons,” *Phys. Rev. D* **114**, 034056 (2026). [arXiv:2605.17389](#)

Preprints / Under Review

- [24] W.-C. Chang, C.-Y. Hsieh and **S. Puhan**, “Leading-neutron electroproduction at HERA and the EIC: Sullivan process, target fragmentation, and pion PDFs,” [arXiv:2609.03368](#)
- [25] **S. Puhan**, S. Sharma and N. Kumar, “Extraction of pion unpolarized quark and gluon generalized parton distributions using deep neural-networks,” [arXiv:2608.07085](#)

- [26] A. Dwibedi, **S. Puhana** and S. Ghosh, “Mechanical distribution of the pseudoscalar charmonium and bottomonium on the light-front,” [arXiv:2606.07073](#)
- [27] S. Sharma, A. Aparin, **S. Puhana**, N. Kumar and H. Dahiya, “Moderate-to-large- x gluon helicity from J/ψ production at $\sqrt{s} = 27$ GeV,” [arXiv:2605.30945](#)
- [28] **S. Puhana**, S. Sharma, N. Kumar and H. Dahiya, “Extraction of pion unpolarized quark generalized parton distribution from charge form factors,” *Communicated to Phys. Rev. D.* [arXiv:2604.11116](#)
- [29] A. Dwibedi, **S. Puhana**, S. Ghosh and H. Dahiya, “Distribution functions of radially excited pion using the light-front quark model,” *Communicated to Phys. Lett. B.* [arXiv:2601.06628](#)
- [30] A. Yadav, **S. Puhana** and H. Dahiya, “Radiative transitions for the ground and excited charmonia states,” *Submitted to Phys. Rev. D.* [arXiv:2504.14864](#)

Conference Proceedings.....

- [C1] **S. Puhana** and H. Dahiya, “Spatial and transverse structure of heavy B- and D-mesons,” *PoS HQL2023*, 089 (2024), DOI: 10.22323/1.462.0089. [arXiv:2408.07717](#)
- [C2] **S. Puhana**, N. Kumar and H. Dahiya, “Understanding photon TMDs using light-front wave function,” *Hadron Physics (HP 2024) Proceedings.* [arXiv:2504.01419](#)
- [C3] **S. Puhana**, N. Kumar and H. Dahiya, “Spin-1 unpolarized GPDs in light front formalism,” *DAE Symp. Nucl. Phys.* **68** (2025) 825.
- [C4] A. Yadav, **S. Puhana** and H. Dahiya, “Radiative transition of $\psi(2S) \rightarrow \chi_{c0}(1P) + \gamma$ in light-front quark model,” *DAE Symp. Nucl. Phys.* **68** (2025) 855.
- [C5] A. Gautam, D. Singh, **S. Puhana**, N. Kaur, A. Kumar, H. Dahiya and S. Dutt, “In-medium properties of pseudoscalar mesons in symmetric nuclear matter,” *DAE Symp. Nucl. Phys.* **68** (2025) 851.
- [C6] R. Acharyya, **S. Puhana**, N. Kumar and H. Dahiya, “Spectroscopy of excited states of D^+ and D_s^+ mesons in the light-front quark model,” *68th DAE Nuclear Physics Symposium 2024*, IIT Roorkee.
- [C7] **S. Puhana** and H. Dahiya, “Valence quark structure of pion using light front dynamics,” *Springer Proc. Phys.* **432**, 650 (2026) [XXVI DAE-BRNS HEP Symposium 2024].
- [C8] A. Gautam, D. Singh, **S. Puhana**, N. Kaur, A. Kumar, H. Dahiya and S. Dutt, “Behavior of pion in asymmetric nuclear medium,” *Springer Proc. Phys.* **432**, 978 (2026) [XXVI DAE-BRNS HEP Symposium 2024].
- [C9] A. Kumar, D. Singh, **S. Puhana**, N. Kaur, M. Kaur, S. Dutt and H. Dahiya, “Kaons structure in dense nuclear medium,” *Springer Proc. Phys.* **432**, 1111 (2026) [XXVI DAE-BRNS HEP Symposium 2024].
- [C10] **S. Puhana** and H. Dahiya, “Leading-twist TMDs for pion and kaon using power law wave functions,” *ATHICS 2025 Proceedings.*
- [C11] **S. Puhana** and H. Dahiya, “Extraction of pion parton distribution function using neural network,” *DAE Symp. Nucl. Phys.* **69** (2026) 849.
- [C12] **S. Puhana**, S. Sharma, N. Kumar and H. Dahiya, “Pion sub-leading twist generalized transverse momentum dependent parton distributions E_2 in light-front quark model,” *DAE Symp. Nucl. Phys.* **69** (2026) 859.
- [C13] Abhishek K. P, R. Pandey, **S. Puhana**, S. Dutt, H. Dahiya and A. Kumar, “Extraction of unpolarized quark TMDs of proton from FNAL-E-0605 data using iminuit,” *DAE Symp. Nucl. Phys.* **69** (2026) 877.
- [C14] S. Sharma, **S. Puhana**, N. Kumar and H. Dahiya, “Model-based constraints on proton TMDs,” *DAE Symp. Nucl. Phys.* **69** (2026) 897.
- [C15] Hari Govind P, R. Pandey, **S. Puhana**, S. Dutt, H. Dahiya and A. Kumar, “Extraction of $f_1(x, k_\perp^2)$ quark TMDs of proton from E288 Drell–Yan data,” *DAE Symp. Nucl. Phys.* **69** (2026) 963.

Invited Talks & Seminars

Prof. C. R. Ji Research Group Seminar, NC State University, USA

Extraction of Pion GPDs

17 Apr 2026 (online)

Discussed data-driven extraction of pion unpolarized quark GPDs at zero skewness, form factor fitting, and impact-parameter-space interpretations.

BLTP, JINR, Dubna, Russia

Gravitational Form Factors of Pion

Aug 2025

Conference: “Interplay of Gravity and Strong Interaction.”

Prof. C. R. Ji Research Group Seminar, NC State University, USA

Internal Structure of Light Mesons Using Distribution Functions

4 Jul 2025 (online)

JINR, Dubna, Russia

Internal Structure of Pion, Kaon, and ρ Meson in Light-Front Dynamics

28 Mar 2025 (online)

Covered TMDs, GPDs, form factors, PDFs, and medium modifications.

68th DAE SNP Symposium, IIT Roorkee

Spin-1 Unpolarized GPDs in Light-Front Formalism

December 2024

HHIQCD2024, Yukawa Institute for Theoretical Physics, Kyoto University, Japan

Structural Analysis of Pion and Kaon with Distribution Functions

Oct–Nov 2024

EIC Workshop, ICTS-TIFR, Bengaluru

Structural Analysis of Pion with Distribution Functions (TMDs, GPDs, GDAs)

Jan–Feb 2024

Workshops & Conferences

Institute of Physics, Academia Sinica, Taipei, Taiwan

EIC-Asia Workshop on QCD and Hadron Structure

Apr–May 2026

Topics included lattice QCD, meson and nucleon structures at the EIC.

Indian Institute of Technology Bombay, Mumbai

2026 IITB-CFNS-CTEQ School on Perturbative QCD for EIC

Feb 2026

Curriculum: perturbative QCD, recent experimental results, hands-on Monte Carlo tutorials.

ICTS, Bengaluru

School for Advanced Topics in Particle Physics (SATPP)

Jan 2026

Selected Topics in Effective Field Theories (EFTs): low-energy and high-energy EFTs.

IIT Bhilai, Chhattisgarh

International Conference on Frontiers of High Energy Physics (ICFHEP 2025)

Feb 2025

Yukawa Institute for Theoretical Physics, Kyoto University, Japan

Hadrons and Hadron Interactions in QCD (HHIQCD2024) [talk]

Oct–Nov 2024

Lyceum Alpinum, Zuoz, Switzerland [partial support from PSI]

PSI Particle Physics Summer School: From Low to High

Aug 2024

ICTS-TIFR, Bengaluru [full support from ICTS]

International School and Workshop on Hadron Structure at the EIC [talk]

Jan–Feb 2024

TIFR, Mumbai [full support from TIFR]

16th International Conference on Heavy Quarks and Leptons (HQL 2023) [poster]

Nov–Dec 2023

IISER Mohali

XXV DAE-BRNS High Energy Physics Symposium

Dec 2022

NIT Jalandhar

HADRON PHYSICS 2023 [workshop]

Jan 2023

IIT Mandi and NIT Jalandhar

Shivalik HEPCATs meetings

Jan & May 2023

TIFR, Mumbai

Celebrating a Decade of the Higgs [webinar]

Jun 2022 (online)

Institute of Physics, Bhubaneswar

International Meeting on Effective Pathways to New Physics (IMEPNP)

Feb 2022 (online)

Research Awards & Grants

Awards.....

Jan 2025: Premier Research Award in the Department of Physics, NIT Jalandhar (26th January 2025).

Financial Support.....

JINR, Dubna, Russia

Support

Jun-Sep 2025

For Visiting Fellowship at BLTP, JINR under Prof. Oleg V. Teryaev.

Yukawa Institute for Theoretical Physics (YITP), Kyoto University, Japan

Partial Travel Support

Oct-Nov 2024

For participation in HHIQCD2024.

Paul Scherrer Institute (PSI), Switzerland

Partial Travel Support

Aug 2024

For participation in PSI Particle Physics Summer School, Zuz.uz.

International Centre for Theoretical Sciences (ICTS), Bengaluru

Full Support

Jan-Feb 2024

For participation in the EIC School and Workshop.

Tata Institute of Fundamental Research (TIFR), Mumbai

Full Support

Nov-Dec 2023

For participation in HQL 2023.

Achievements

- AIR 209 in JEST (Physics), 2021
- AIR 1544 in GATE (Physics), 2021
- AIR 877 in JAM (Physics), 2019
- Qualified JEE, 2016
- 2nd Topper in Physics, CHSE Odisha (state-level), 2016
- NTSE Scholar, 2013
- 2nd Rank, District Debate, Food Supply Dept. Odisha, 2010
- 3rd Rank, Jhadeswar Science Residential School, 2014
- 2nd Rank, District Exam, Siddhi Vinayak Residential Science College, Balasore, 2014

Technical Skills

Programming: Mathematica (Expert), Python / NumPy / SciPy (Expert), C++ (Intermediate), Git / Linux

AI / ML: Machine Learning and Neural Networks (active use in research), Monte Carlo simulations (Pythia 8), Quantum Computing (learning)

Phenomenology: LHAPDF, xFitter, DGLAP / CSS evolution, iminuit

Typesetting: L^AT_EX, Overleaf, Scientific Writing

Software: Mathematica, Jupyter, Microsoft Office

Operating Systems: Windows, Linux

Web: HTML (learning)